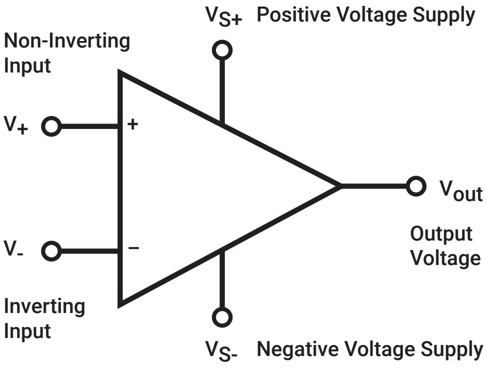
# OP AMPS

## Op Amps

Operational amplifiers, or op amps, are one of the fundamental building blocks of many electronic designs. Uses can range from simply making a small signal larger to advanced analog signal processing.




### Op Amp Types

| Type | Description |
|---|---|
| Buffer | Pre-configured to be used as an analog buffer, normally with unity gain |
| Current Sense | Output voltage proportional to current passed through sense resistor |
| Differential | A differential is designed to amplify the difference between two signals |
| Instrumentation | Differential amplifier with buffered inputs and single resistor gain adjustment |
| Power | An op amp with an output power stage allowing it to source more current |
| Programmable Gain | Op amps with variable gain that can be programmed digitally |
| Transconductance | An amplifier that takes a voltage input and produces a current output |
| Transimpedance | An amplifier that takes a current input and produces a voltage output |
| Audio | Amplifies signal adding minimal noise and low distortion for audio applications |

### Common Terms

| Term | Definition |
|---|---|
| Gain | Av is determined by Rf (feedback Resistor) / Rin (Input Resistor) |
| Offset Null | Offset voltage needed to bring the output to zero volts |
| Slew Rate | Maximum rate of change of output per unit of time |
| CMRR | Common Mode Rejection Ratio |
| Single Supply | Requires only ground and a positive voltage supply |
| Dual Supply | Requires positive and negative supply voltages |
| Inverting | Amplifies a signal with a negative output |
| Non-Inverting | Amplified a signal with a positive output |